Vinayak Kamat

AI/ML Engineer | LLM Agents, RAG & Production ML

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Summary

AI/ML Engineer with 1+ year of production experience building LLM-powered systems. Hands-on with LangChain/LlamaIndex, RAG pipelines, and LLM evaluation. Built Aethelgard, a real-time hallucination detection gateway scoring LLM outputs via FastAPI and Datadog. Comfortable deploying on AWS and optimizing inference (PyTorch-to-ONNX). MSc Financial Engineering candidate with domain exposure to financial data workflows.

Skills

LLM & Agent Frameworks: LangChain, LangGraph, LlamaIndex, OpenAI API, Anthropic Claude API, Prompt Engineering (Few-shot, CoT, ReAct), Function Calling, Structured JSON Outputs

RAG & Vector Databases: RAG pipelines, Chroma, FAISS, Embeddings, Retrieval evaluation, Chunking strategies

Programming: Python, C++, JavaScript, SQL, FastAPI, Flask, REST APIs

ML & Deep Learning: PyTorch, Transformers, OpenCV, spaCy, ONNX, Model Calibration

MLOps & Cloud: MLflow, Docker, Git, AWS (Lambda, S3, ECR, DynamoDB), CI/CD, Datadog, Model Monitoring, Drift Detection

Data: Pandas, NumPy

Experience

KAN Innovations, AI/ML Engineer — Mumbai, Maharashtra — May 2025 – present — 1 year 7 months

- Built and evaluated production RAG pipelines using LangChain/LlamaIndex and Chroma, implementing retrieval quality checks and output validation to measure accuracy
- Implemented function calling and structured JSON output validation for LLM-driven data extraction from financial documents
- Deployed ML services on AWS (Lambda, ECR, DynamoDB); optimized PyTorch-to-ONNX inference for latency reduction
- Used MLflow for experiment tracking, model versioning, and reproducible workflows

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra — Feb 2025 – May 2025 — 3 months

- Built a PyTorch-based detection model achieving ~85% accuracy, from dataset preparation through training and evaluation
- Improved an existing posture detection model through data and training refinements

Projects

Aethelgard — LLM Evaluation & Guardrails

Real-time gateway that grades LLM outputs for hallucinations and risk, with observability.

- Built a FastAPI gateway using a secondary model for semantic risk scoring, with metrics streamed to Datadog for continuous monitoring
- Designed automated evaluation logic and fallback mechanisms so low-confidence outputs degrade safely before reaching users
- Implemented structured output validation to catch malformed or unreliable LLM responses

LangGraph Agent Demo — Tool Use & Structured Extraction
Multi-step agentic workflow demonstrating tool use and structured JSON extraction.

- Built a LangGraph-based agent that routes queries, calls mock APIs (invoice/transaction lookup), and returns validated JSON outputs
- Implemented few-shot prompting and ReAct-style reasoning for reliable tool selection
- Added fallback handling for tool-call failures to keep the agent reliable

Financial AI Agent (Demo)
Prototype agent for financial data retrieval and reasoning using Python and LLMs.

- Built modular retrieval and reasoning pipelines for financial insights
- Implemented query routing and response refinement with structured output parsing

Education

MSc **WorldQuant University**, Financial Engineering — Remote, Washington DC
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

Achievements

Qualified GATE (Graduate Aptitude Test in Engineering)

Active open-source contributor with AI/ML projects on GitHub